Skilled Nursing Facility Prototype Scorecard Spreadsheet Review

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January 14, 2026

Summary of Major Comments

The following comments represent errors found when comparing values from the scorecard to the source cited in the scorecard:

- Wrong roof requirement in CZ10 (NURS36-Envelope)
- SHGC wrong in CZ3 to CZ9 (NURS38-Envelope)
- Corridor_2 water heater use (NURS73-Water Heater)
- Lighting power error incorrect in some spaces (NURS82-Interior Lights, NURS83-Interior Lights, NURS85-Interior Lights, NURS87-Interior Lights, NURS91-Interior Lights)
- Plug load power density is incorrect for some spaces (NURS100-Equipment)

The following are the highest priority comments:

- Different vintages of skilled nursing facilities look very different, and no modern one looks like the prototype (NURS12-Prototype, NURS20-Zones)
- Area of building uncertain due to conflicting information (NURS14-Prototype)
- Window-to-wall ratio uncertain due to conflicting information (NURSE16-Prototype)
- Two occupants per patient room are unusual in recent skilled nursing facilities (NURS22-Zones)
- Patient room size is too small (NURS23-Zones)
- Each patient room should have a private bathroom (NURS24-Zones)
- Number of staff too high (NURS30-Zones)
- The space type Laundary_1 is shown with a space/thermal zone of BATH_Core_ZN_1 (NURS31-Zones)
- Some patient rooms are associated with the wrong HVAC zones (NURS32-Zones)
- Infiltration values cannot be reproduced due to area values (NURS44-Envelope)
- EER for 2008-2018 vintage may be wrong (NURS48-HVAC System)

- Water heater capacity is not based on the sum of space loads (NURS75-Water Heater)
- Water heater storage volume calculated incorrectly (NURSE77-Water Heater)
- New construction lighting power not using Aging Eye/Low-vision values (NURS81-Interior Lights)
- No elevator loads described (NURS104-Equipment)
- Schedules could not be verified because the source was unclear (NURS106-Schedules)

The following comments represent other important issues found:

- The Notes tab does not justify many assumptions like other scorecards (NURS8-Notes)
- Need to identify the type of skilled nursing facility (NURS10-Notes)
- Need to identify if facility is urban or suburban (NURSE11-Notes)
- The naming used in the diagram does not match the list of spaces (NURS26-Zones)
- Many types of spaces are missing from the program (NURSE34-Zones)
- SHGC to VT conversion factor for 1998-2007 vintage not typical (NURS39-Envelope)
- Similar rooms have inconsistent ventilation rate (NURS58-Ventilation, NURS60-Ventilation, NURS61-Ventilation, NURSE62-Ventilation, NURS63-Ventilation)
- Patient room thermostat settings are insufficiently diverse (NURS70-Thermostat)
- Water main temperature not being used for sizing (NURS76-Water Heater)
- Additional power for ornamental lighting allowed for some existing spaces (NURS93-Interior Lights)
- No outdoor illuminated spaces assumed for patients (NURS95-Ext Lights)
- Façade lighting uses a perimeter value inconsistent with others (NURS96-Ext Lights)
- Weird high corridor occupancy from 8 pm to 10 pm (NURS105-Schedule)

In addition, the following comments describe the lack of disclosed assumptions that make verifying the values in the scorecard nearly impossible:

- Which HCAI and CalHHS resources were actually used? (NURS4-Notes, NURS5-Notes)
- What is the source of the construction rate? (NURS7-Notes)
- Why two stories? (NURS15-Prototype)
- How many beds? (NURS21-Zones)
- What is the function of special-purpose rooms? (NURS29-Zones, NURS56-Ventilation, NURS88-Interior Lights)
- Which footnotes are related to specific values? (NURS33-Zones)

- How are the wall u-factors for 1978-1997 vintage determined? (NURS35-Envelope)
- Justification for weighting single and double pane is missing (NURS40-Envelope)
- How was the HVAC system type chosen for new construction? (NURS45-HVAC)
- Explain how 2008-2018 HVAC efficiency values were chosen (NURS49-HVAC)
- How is the airflow rate chosen to pick the right pressure drop assumption? (NURS50-HVAC) or power per airflow (NURS52-HVAC)
- Please identify which source is used for each ventilation rate (NURS57-Ventilation)
- Why is pharmacy not using the Standard 170 ventilation rate? (NURS59-Ventilation)
- Where do the exhaust rates come from? (NURS64-Ventilation)
- What is the basis for the amount of kitchen exhaust hood? (NURS67-Ventilation, NURS68-Ventilation)
- What is the specific source of the thermostat setpoints? (NURS69-Thermostat)
- What category is used to classify the water heating use for some spaces? (NURS71-Water Heater)
- Why use Title 24-2016 for existing lighting when other scorecards use 2013 or 2019? (NURS80-Interior Lights)
- What lighting space type is assumed for some spaces? (NURS86-Interior Lights, NURS92-Interior Lights)
- How was the size of the parking lot determined? (NURS94-Ext Lights)
- What space type was assumed for plug loads for some spaces? (NURSE103-Equipment)

Background

The review of the "score card," which is a spreadsheet summarizing the model assumptions, is broken down by spreadsheet tab, which includes:

- Notes
- Prototype
- Zones
- Envelope
- HVAC System
- Ventilation
- Thermostat
- Water Heater
- Interior Lighting
- Ext Lighting

- Equipment
- Schedules
- Energy Usage
- Weights
- PV
- EV Charger

The comments on these tabs are labeled using the following format NURSx-<TAB>-<CELLRANGE>), where <TAB> is the name of the tab in the spreadsheet and <CELLRANGE> indicates the specific cells. Note that no tabs were reviewed after the Schedules tab.

A copy of the spreadsheet is available that includes which cells on each tab were checked against the sources and were consistent (shown in light green) and which had anomalies and are described in a comment below (shown in tan).

NURS1-Notes (n/a)

Rather than repeat it, please see comment PPD1 in my single-family residential prototype review, LHSC1-Notes in the large hotel review, and SHSC1-Notes in the small hotel review.

NURS2-Notes (n/a)

Rather than repeat it, please see comment PPD2 in my single-family residential prototype review, LHSC2-Notes in the large hotel review, and SHSC2-Notes in the small hotel review.

NURS3-Notes (n/a)

To provide a thorough review of the skilled nursing prototype, I reviewed several great resources shown below prior to beginning my review. Originally, over 50 possible resources were identified, but the following seemed the most important.

Sources that *should* be reviewed for background information and are fundamental for reviewing the nursing home prototype are the following:

- *ASHRAE Handbook HVAC Applications Chapter 9 "Health Care Facilities"*
  Essential for understanding healthcare HVAC. If only one thing is reviewed, this should be it.

https://store.accuristech.com/standards/2023-ashrae-handbook-hvac-applications-i-p?product_id=2225673

- *Building Type Basics for Senior Living - Second Edition*
  Many specifics about rooms and styles of facilities, as well as providing numbers for room sizes.
  https://www.wiley.com/en-us/Building+Type+Basics+for+Senior+Living%2C+2nd+Edition-p-9781118007457
- *Polytrauma Rehabilitation Center Design Guide - U.S. Veterans Affairs*
  Many detailed room layouts for each space type are provided in both inpatient nursing units and transitional rehab.
  https://vatilms.va.gov/vatilms/reports/PG-18-12
- *Small House (SH) Model Design Guide - U.S. Veterans Affairs*
  Great reference for a small house facility to provide skilled nursing in a residential-like setting.
  https://vatilms.va.gov/vatilms/reports/PG-18-12
- *Guidelines for Design and Construction of Residential Health, Care, and Support Facilities*
  Many details on specific areas of residential health care. Almost too much information.
  https://fgiguidelines.org/
- *Green House Project by Aging IN*
  Information about the small house approach.
  https://thegreenhouseproject.org/
- *Design Guide for Long Term Care Homes*
  Good description of space programming
  https://www.fgiguidelines.org/wp-content/uploads/2018/03/MMP_DesignGuideLongTermCareHomes_2018.01.pdf
- *On the Energy Performance Design of a Skilled Nursing Facility Building*
  Description of an EnergyPlus model for an SNF, including setpoints and types of spaces and details of the HVAC system and internal gains for different spaces.
  https://docs.lib.purdue.edu/cgi/viewcontent.cgi?params=/context/ihpbc/article/1118/&path_info=3365.pdf
- *ANSI/ASHRAE/ASHE Standard 170-2021 Ventilation of Health Care Facilities*
  Contains a table of design parameters related to ventilation
  https://store.accuristech.com/standards/ashrae-170-2021?product_id=2212971

Additional useful sources could be added to the list shown in the scorecard spreadsheet:

- (Re)designing the Continuum of Care for Older Adults https://link.springer.com/book/10.1007/978-3-031-20970-3
- Methodology for Modeling Building Energy Performance across the Commercial Sector https://docs.nrel.gov/docs/fy08osti/41956.pdf
- A Look at Nursing Facility Characteristics in 2025 https://www.kff.org/medicaid/a-look-at-nursing-facility-characteristics/
- Building energy performance simulation model for a nearly zero-energy nursing home in Belgium https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/JSZRNT
- Building Type Basics for Healthcare Facilities 2nd Ed https://www.bnibooks.com/products/building-type-basics-for-healthcare-facilities-second-edition?_pos=2&_sid=92a3e800a&_ss=r
- Design Details for Health: Making the Most of Design's Healing Potential, 2nd Edition https://www.wiley.com/en-us/Design+Details+for+Health%3A+Making+the+Most+of+Design's+Healing+Potential%2C+2nd+Edition-p-9780470926840
- Health Care Facilities – Whole Building Design Guide https://www.wbdg.org/building-types/health-care-facilities
- Nexus of Thermal Resilience and Energy Efficiency in Buildings: A case study of a nursing home https://doi.org/10.1016/j.buildenv.2020.106842
- Nursing Home - Whole Building Design Guide https://www.wbdg.org/building-types/health-care-facilities/nursing-home
- Nursing Home Characteristics and Resident Quality of Care Outcomes: A Scoping Review of Recent Empirical Research https://doi.org/10.31389/jltc.235
- Paying Up For Square Footage In Skilled Nursing https://seniorcare.levinassociates.com/2017/07/03/paying-square-footage-skilled-nursing/
- ENERGY STAR Score for Senior Care Communities in the United States https://www.energystar.gov/sites/default/files/tools/Senior_Living_April_2021_EN_508.pdf

- Turning Nursing Homes into True Homes: The Household Model https://www.chcs.org/turning-nursing-homes-into-true-homes-the-household-model/
- Wikipedia https://en.wikipedia.org/wiki/Nursing_home https://en.wikipedia.org/wiki/Nursing_home_care_in_the_United_States

By utilizing additional sources of information, the validity of assumptions can be strengthened. Some parameters may be California-specific, but comparing assumptions across multiple sources lends credibility to any assumptions made, even when a California-specific database ultimately may be the source of the information. Tables showing parameters across multiple sources, and then values chosen being highlighted, can be a powerful expression of when parameters are solidly determined by being similar in multiple data sets versus being based on just a single data set. Please also note that some of the above references may include citations of relevant reports and papers that could be reviewed as further sources.

NURS4-Notes (B2:F2) and others

HCAI is referenced here to a generic web page:

https://hcai.ca.gov/facilities/building-safety/

That page contains links to many different specific resources, but the references in other places in the scorecard to HCAI are not to specific destinations. The unexplained and nonspecific references to HCAI continue in other locations:

- The Prototype tab cell C9, related to the aspect ratio of 3.0 "Matching with HCAI facility drawings," does not indicate which web pages such drawings are located on.
- Envelope tab cells A45:X45, which is a footnote to a table, states in part, "HCAI usually expects Type I-A or I-B for multi-story SNFs, or Type II-A for smaller single-story facilities," does not indicate which table (envelope constructions or infiltration) this note applies to, or what these Types I-A, I-B, or II-A mean or are referencing.
- Ventilation tab cell K9 related to "Food_Preperation_1" space/thermal zone has a note "ASHRAE 170, ASHRAE 62.1 User's Manual Equation 6-1, ASHRAE ventilation of health care facilities, HCAI A6, FGI 2022" which mentions HCAI for A6 without explanation or further reference. This note is repeated for Ventilation A21:L27, covering special-purpose rooms, patient rooms, bathrooms, and laundry without any further explanation.

- Ventilation tab cell A84 is a footnote under a small table related to Hood Appliance mentions "Special Requirements per HCAI for ventilation after Covid" without further explanation or source, or what parameter this note applies to.

NURS5-Notes (B4:F4) and Prototype (C8)

The citation to CalHHS 2025 on the Notes tab is to a web page:

https://data.chhs.ca.gov/dataset/?tags=nursing+homes

which contains links to three different databases. It would be good to specify which one was the source of information.

The only other reference to CalHHS is on the Prototype tab for the Total Floor Area with the note "Supported by CalHHS 2025 data from CDPH (Same as CPUC)" but the abbreviation "CDPH" is not defined, although probably is for the California Department of Public Health. It would be better to explain exactly what data is used and where to find it to justify the total floor area.

NURS6-Notes (B6:F7, B9:F9)

No links are provided for Title 24 versions or ASHRAE 90.1, 170, or 62.1.

NURS7-Notes (A16:B30)

No citation is included for the source of data for the Construction Rate diagram shown, although it might be from CDPH, since that is shown in the heading, or it could be from Dodge data, since that is the source or similar information on other scorecards.

NURS8-Notes (n/a)

In general, other scorecards reviewed so far have had substantially more information on the Notes tab. This notes tab seems to not include information about the construction rate over time, justification of area or number of floors, why representative years were chosen for different vintages were chosen, other graphs supporting major assumptions, alternative data sources to help justify major assumptions, primary energy source, how the categorization of "skilled nursing" is differentiated from other related buildings with slightly different categories. In general, the notes seem weak compared to other scorecards.

NURS9-Notes (B5:F5)

The reference to the DEER prototype is currently to this URL:

https://github.com/sound-data/DEER-Prototypes-EnergyPlus

But it could be more specific to here:

https://github.com/sound-data/DEER-Prototypes-EnergyPlus/tree/main/prototypes/Nrs/templates

NURS10-Notes (n/a)

There are several different categories of skilled nursing facilities that have different focuses and different layouts due to their focus. Categories include:

- Long-term care
- Short-term rehabilitation
- Memory care
- Hospice care

It would be good to identify what the focus of the prototype facility is; it appears that it is none of these categories, and that not understanding the implications of such categories makes the prototype seem less realistic. Short-term rehabilitation facilities, not surprisingly, have more spaces dedicated to physical therapy (similar to small gymnasiums) and occupational therapy (training rooms often include kitchens). Long-term care facilities often try to focus on creating home-like settings and will have common areas that are more like living rooms, dining rooms, open kitchens, dens, community rooms, and chapels.

NURS11-Notes (n/a)

Skilled nursing facilities vary significantly depending on whether they are in urban areas, suburban areas, or rural areas. Similar to other types of buildings, urban locations are often taller and take up less land, and may include parking facilities. The buildings in suburban and rural areas have fewer floors with surrounding, which seems to be the approach for the prototype. This distinction is important, so the building should be described as a "suburban or rural skilled nursing facility."

NURS12-Prototype (n/a)

For most buildings, using the same layout across vintages makes sense since the design of those buildings has changed little over many years. But this assumption is not valid for skilled nursing facilities especially long-term care facilities. Due to the evidence-based care approach, buildings that provide long-term skilled nursing facilities that are being built now look very little like the ones built in the 1970s through 1990s. An understanding the occupants comfort in the environment being an important aspect of their health has transformed the design of these buildings. Instead of appearing to be hospital wards, they are looking like homes suitable for a dozen residents to occupy. Please see the U.S. Department of Veterans Affairs' Small House Design Guide (https://vatilms.va.gov/vatilms/reports/PG-18-12) for a great example of how this is different. This may appear to provide a stumbling block for a prototype, but I would suggest further research as to the era of this transition. Perhaps the distribution of areas should vary parametrically with vintage. The Small House Design Guide includes the statement: "This commitment to cultural transformation was made by VHA in 2005, in keeping with shifting standards in the nursing home industry." Prototypes are intended to represent buildings from different eras, and in this case, layout, types of spaces, and areas devoted to different purposes should be different for the pre-1978 and 1978-1997 vintages from 1998-2007, 2008-2018, and new construction vintages (further research for the year to assume the change is warranted). Doing otherwise will not represent either old or new buildings accurately. Since the purpose of simulating various vintages is not to compare between them but instead to capture the energy impact of various measures applied to all of them, having multiple models representing the evolution of skilled nursing facilities should be acceptable. Also see NURS34-Zones for a list of space types that should be present in a short- or long-term care facility.

NURS13-Prototype (A7:C7)

While a rectangular shape for the building may be supported by CEUC and CEC (please note that specific references that show that support are missing), a model that is long and rectangular does suffer from being sensitive to the orientation of the building. The building loads when the axis of the building runs north, and south is likely to be much more driven by morning and evening solar loads. I would suspect that those same sources probably also show that L-shaped skilled nursing facilities are also common and that shape does not have nearly the sensitivity to orientation, especially when only two stories tall.

NURS14-Prototype (A8:C9)

The total floor area of 30,100 square feet seems to be consistent with the statistics shown on the Weights tab (which should probably be on the Notes tab), indicating an average of 97 beds per facility and 343 square feet per bed. This results in 33,271 square feet, which is about 10% larger than the prototype. While this is close enough for modeling purposes, it is not clear why 33,271 square feet was not used.

The floor area of 30,100 square feet, though, is not consistent with the 29,690 square feet shown on the Zones tab cell F116, which is the sum of the zone areas (except plenums).

The "430 ft x 140 ft" shown in parentheses after the 30,100 ft2 is not consistent since that multiplies out to 60,200 square feet. But those dimensions imply that it is the length and depth of a single floor, and since it is two floors, that would imply the building is 120,400 square feet. This is much larger than the 33,271 square feet justified on the Weights tab.

The aspect ratio of 3.0 is not consistent with the images, which look much longer and thinner, and probably an aspect ratio of 6 or more. Also, the "430 ft x 140 ft" results in a 3.07 aspect ratio, which would round to 3.1, not 3.0. The plan views shown on the Zones tab do look like an aspect ratio of around 3, so it may just be the image in cell B7 on the Prototype tab that does not look correct.

NURS15-Prototype (A10:C10)

A two-floor building for a skilled nursing facility is quite reasonable, and no data other than "Same as CPUC" is provided to justify the value. Because of the requirement of having an elevator (usually oversized to hold a gurney) for this type of building when more than one floor, single-floor skilled nursing facilities are quite common (the only building of this type that I have spent much time in was a single-floor facility). For non-residential buildings in general, once the investment is planned for an elevator, it is often cost-effective to have a three-story building instead of a two-story building since the incremental cost of the elevator itself from two to three stories is small, and the roof and foundation areas are reduced.

NURS16-Prototype (A12:C12)

The 20% window-to-wall ratio is reasonable, but no justification is provided beyond "Same as CPUC"; in addition, the window dimensions provided in parentheses:

Window Dimensions:

107.5 ft x 5.22 ft on the long side of facade

35 ft x 5.22 ft on the short side of the façade

Implies long curtain walls, like in high-rise office buildings. This seems very unlikely for a skilled nursing facility where punched windows in individual rooms are probably more common. Perhaps this is just describing a modeling simplification; if so, it should be described that way.

The total window area described is 1488 square feet, but I think this is intended for each floor, which would result in a total window area of 2975 square feet.

285 ft (total window width per floor) x 5.22 ft (window height) x 2 stories = 2975 square feet

The exterior walls, based on 430 ft x 140 ft and 13 ft floor-to-floor height and two stories, are 29,640 square feet

1140 ft (circumference) x 26 ft (height) = 29,640 square feet.

Given this, the computed window-to-wall ratio is 10%, not 20%.

Also note that the windows described and the windows shown in the figure in cell B7 do not seem consistent since the figure shows them more spread out and different on the first and second floors.

This will impact the envelope requirements from Title 24-2001 since that is dependent on the window-to-wall ratio.

NURS17-Prototype (A13:C13)

Some shading in the form of the main entrance overhang is very common in buildings where ambulances or other vehicles are likely to be transporting people with serious medical issues who may not be able to walk or be on a gurney.

NURS18-Prototype (A16:C16)

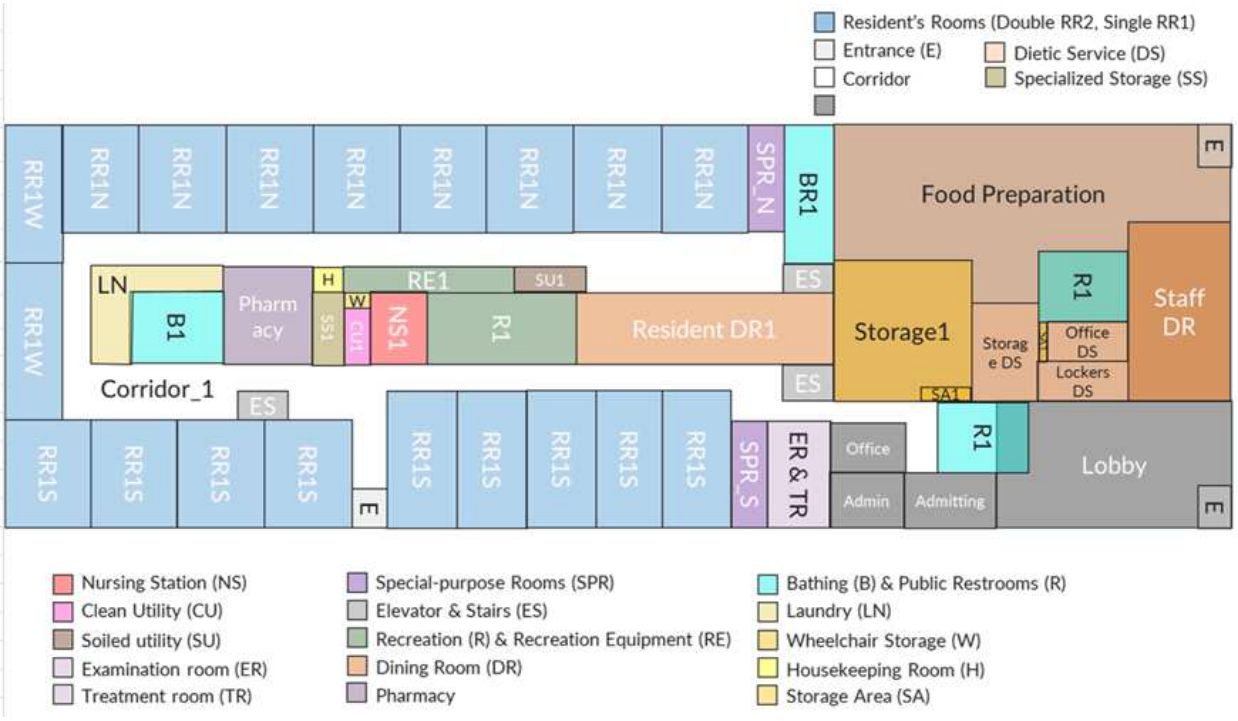
The glazing sill height of 3 ft seems reasonable, but the parenthetical statement after that, "(top of the window is 4.78 ft high with 5.22 ft high glass)" seems confusing. A window height may include a frame, but in that case, the window would be the larger height compared to the glass, not the smaller height.

NURS19-Prototype (A28:D64)

It would be great if multiple prototypes were being compared here, but apparently, only the CPUC prototype exists. This comparison table should be removed, or additional building prototypes should be located for the sake of comparison.

NURS20-Zones (n/a)

Since the category "skilled nursing" encompasses both short-term and long-term care as well as memory and hospice care (see NURS10-Notes), it is difficult to evaluate the zones presented on the Zones tab. This is further complicated by the evolution of facilities during the vintages being evaluated from hospital-like wards to home-like settings (see NURS12-Prototype). By examining the layouts presented on the Zones tab, it appears that this is not a short-term care facility due to the lack of rehabilitation rooms, so the model seems to be a long-term care facility. In addition, it is laid out as an old-fashioned hospital ward, so the assumption must be that this is a long-term care facility requiring skilled nursing that was built like it was in the 1970s to 1990s. Since it is two floors, it is likely to be in a suburban or rural setting (see NURS11-Notes). These assumptions will generally be used in the rest of the comments, although some comments may suggest alternatives if other assumptions for the building type, location, and style are different.



NURS21-Zones (A24:I42, A84:I113)

The number of beds is a critical metric for short and long-term care facilities since it represents the number of patients. The Weights tab (see NURS108-Weights) shows 97 beds as the average. The prototype tab does not mention a number of patients or beds, but the Zones tab shows 37 people on the first floor and 60 people on the second floor in the spaces and zones described as "Patient," matching the 97. Given the importance of the number of beds as a metric in all inpatient health facilities, it should probably appear on the Prototype tab.

NURS22-Zones (A24:I42, A84:I113)

Of the 49 patient rooms, all have two occupants except for one. Most modern skilled nursing facilities appear to have a single occupant per room. Double occupancy would have been more common in the 1970s-1990s. Older existing long-term care facilities are likely to have retrofitted to have one patient per room.

NURS23-Zones (A24:F42, A84:F113)

The area of each patient room is 280 sqft with two occupants (except the one patient room with a single occupant, which is 207 sqft). Cell I24 includes a note "Complying with T24 part 2 section 1225.5.1.2.1 number of people per area," and the text being referenced is:

1225.5.1.2.1 Patient bedrooms.

Patients shall be accommodated only in rooms with the following minimum floor area, exclusive of toilet rooms, wardrobes, entrance vestibules and fixed furnishings or equipment.

1. Single-patient rooms: 110 square feet (10.21 m^2).
2. Multi-patient rooms: 80 square feet (7.43 m^2) per bed.

The "Building Type Basics for Senior Living" indicates that 290-320 sqft for a private room with a bathroom is more typical. The VA documents suggest 330 sqft per patient room for rehab facilities without a bathroom and 230 sqft per patient private room in long-term care, not including a bathroom. While the 280 sqft for two occupants is within the code limit, it is unlikely that a room that small would hold two occupants. In this case, the code minimum is not governing the practice of design today.

NURS24-Zones (A43)

A common patient bathroom is very unusual. Instead, bathrooms that are private to each room are the most common approach used in at least the last few decades. I believe this is due to several reasons, including privacy, dignity, disease spread, and better health outcomes (based on evidence-based design). None of the patient rooms show required exhaust on the Ventilation tab, so we can assume they don't have bathrooms. The use of a private bathroom that is compatible with wheelchairs typically adds 75-90 sqft for each room.

NURS25-Zones (A2:D2)

It is unclear what the exact meaning of the columns "Space Type", "Space/Thermal Zone", and "HVAC Zone" are. The column "Space Type" does not always seem to represent a “type” of space and instead seems to be a specific space. The “space type” values usually have a "_1" or "_2" after the text, which is related to the floor that they are on. In addition, the “space types” often show a direction such as "_N", "_W", or "_S". Typically, a space type is a category of space rather than a specifically named space. Perhaps the columns should be renamed "Space Name." The HVAC Zone column seems to be the HVAC system name. The "Space/Thermal Zone" seems to be simply the thermal zone name.

NURS26-Zones (n/a)

The naming used on the diagrams and the naming used in the columns "Space Type", "Space/Thermal Zone", and "HVAC Zone" are not consistent, and no mapping is done to clarify how they are related. I would suggest that the diagram names be used in the columns of the table, or additional columns should be added that indicate the names used in the diagrams so that the zone properties can be understood on the diagram.

NURS27-Zones (I3)

The lobby is shown to contain 28 people, which seems very high in a facility with 97 beds, presuming that these are visitors. Most visitors would probably be in patient rooms, and the number of visitors to patients seems too high, also. The 28 people in the lobby might be reasonable if we are presuming that they are some of the residents themselves.

NURS28-Zones (A9:D15)

Kitchen HVAC systems are often dedicated to just kitchens, but the "FoodPrep_HVAC_1" HVAC zone also services the Staff_Dining_1, Office_Dietic_Services_1, Lockers_Staff_1, and Restroom_Dietic_Services_1.

NURS29-Zones (A22:D23)

The Special_purpose_N_1 and Special_purpose_S_1 spaces seem to be treated as more patient rooms and are in HVAC Zones Patient_N_HVAC_1_1 and Patient_S_HVAC_1. The HVAC Zone names may be wrong.

NURS30-Zones (Column I)

The total number of people in the facility is 272, and presuming that 97 people are patients, that implies 175 staff members for 97 patients. This ratio is way too high. The schedules show 80% to 100% occupancy during some hours for almost all zones at the same time, except corridors, which have an occupancy of 22 people. The full-time equivalent (FTE) staff for residents is usually near 0.50 (see the following table from Building Type Basics for Senior Living page 326), and since staff is present 24x7, this would be spread over multiple shifts, implying that for 97 patients, no more than 30 to 40 other personnel are probably present at most and at night, significantly fewer.

Table 18-1 FTEs per Resident by Labor Department and Property Type

	Independent Living			Assisted Living			CCRC		
Department	Lower Quartile	Median	Upper Quartile	Lower Quartile	Median	Upper Quartile	Lower Quartile	Median	Upper Quartile
Administrative	0.02	0.03	0.05	0.03	0.04	0.05	0.01	0.02	0.03
Dietary	0.08	0.10	0.12	0.06	0.10	0.13	0.11	0.12	0.15
Housekeeping	0.03	0.03	0.03	0.02	0.03	0.04	0.04	0.06	0.07
Maintenance	0.01	0.01	0.02	0.01	0.02	0.02	0.02	0.03	0.05
Marketing	0.01	0.01	0.02	0.01	0.01	0.02	0.01	0.01	0.01
Assisted Living	0.06	0.08	0.12	0.19	0.28	0.37	0.03	0.04	0.08
Skilled Nursing	***	***	***	0.03	0.04	0.09	0.11	0.13	0.21
Other	0.00	0.02	0.03	0.01	0.02	0.03	0.02	0.05	0.09
Total	0.16	0.24	0.32	0.41	0.50	0.62	0.39	0.48	0.60

*** A minority of independent living properties reported nursing-related revenue in properties without licensed skilled nursing units, which is likely indicative of the provision of temporary private-duty nursing services to independent living residents.

Source: American Seniors Housing Association, 2010, p. 63.

NURS31-Zones (A44:D44)

The space type Laundary_1 is shown with a space/thermal zone of BATH_Core_ZN_1, but the HVAC Zone is Service_HVAC_1. It seems like the BATH_Core_ZN_1 is inconsistent with the other descriptions.

NURS32-Zones (A96:D113)

The rows near the bottom of the table show Patient_Rooms in the east, south, and western directions being associated with HVAC zones named for the north direction.

NURS33-Zones (A119:A124)

The large footnote would be more helpful if broken up into individual footnotes that are referenced for each item on the table that uses that source. It is difficult to know what each reference is actually used for. Obviously, not all those sources are used for every value in the table.

NURS34-Zones (n/a)

Types of spaces missing from the program on the Zone tab include

- Resident's open kitchen and pantry
- Resident
- Medication room
- Multipurpose room
- Coffee shop/snack bar
- Gift shop
- Living room
- Library
- Gym
- Art/activity spaces
- Rehabilitation facilities
- Family areas
- Mechanical/electrical rooms
- Staff training rooms

- Counseling
- Testing rooms
- Assistive technology
- Speech language pathology room
- Garage
- Theater
- Chapel

These spaces are typically present in a modern long-term or short-term facility. It is not surprising given the older style "hospital ward" style of skilled nursing facility assumed (see NURS12-Prototype) but that means the prototype is not realistic for present day facilities or even for facilities after 1990. For another list, see Building Type Basics for Senior Living, Appendix I for a sample program and the VA sources for details of different rooms and a list of types of rooms.

NURS35-Envelope (H4)

For the external wall u-factors for the vintage 1978 to 1997, the Title 24-1986 version is referenced and shown as "Section 160.1 – Mandatory Requirements for Building Envelopes," but there is no Section 160.1 in that version of Title 24.

The values shown in the table for u-factors are 0.151 for CZ1 to CZ16. The reciprocal of that value is 6.63, which does show up in a low-rise office table (Table 2-53V1) for packages A and B in climate zone 1. It is really not clear where these values come from, and they cannot be confirmed.

NURS36-Envelope (R8:R9)

The roof requirements for the "pre 1978" and "1978-1997" vintages are shown generally as the same as the values in the "1998-2007" vintage, with the note "Considering 30 years for roof/ insulation replacement." This is consistent except for CZ10, which showed 0.057 for the "1998-2007" vintage value and 0.78 for the "pre 1978" and "1978-1997" vintages. Since the values in other climate zones are 0.78, this is likely a copy-paste error.

NURS37-Envelope (I13:X14)

Repeat of comment SHSC32 - use of 0.338 instead of 0.33 as weighting.

NURS38-Envelope (H20:X20)

On the prototype tab, the window-to-wall ratio is shown as 20% (note that in NURS16-Prototype, when this is recomputed, it seemed like 10%), but for the 1998 to 1997 vintage, the Title 24-2001 is used, and the note says "Table 1-H (Non North, 20-30% total WWR)." The actual table in that version of the standard uses window-to-wall ratio ranges of 0-10%, 11-20%, 21-30%, and 31-40%. Since the prototype is intended to be 20% window-to-wall ratio, the proper range to use would be 11-20%. This impacts the SHGC values of CZ3 to CZ9, which are shown as 0.41 or 0.39 and should be 0.55 or 0.61.

TABLE 1-H—PRESCRIPTIVE ENVELOPE CRITERIA FOR NONRESIDENTIAL BUILDINGS
(Except high-rise residential buildings and guest rooms of hotel/motel buildings)

	CLIMATE ZONES									
	1, 16		~~2~~3-5		6-~~10~~9		2, 10~~11~~-13		14, 15	
Roof/Ceiling										
R-value or	19		19		11		19		19	
~~U-value~~U-factor	0.057		0.057		0.078		0.057		0.057	
Wall										
R-value or	13		11		11		13		13	
~~U-value~~U-factor										
Wood frame	0.084		0.092		0.092		0.084		0.084	
Metal frame	0.182		0.189		0.189		0.182		0.182	
Mass/7.0≤ HC<15.0	0.340		0.430		0.430		0.430		0.430	
Mass/15.0≤HC	0.360		0.650		0.690		0.650		0.400	
Other	0.084		0.092		0.092		0.084		0.084	
Floor/Soffit										
R-value or	19		11		11		11		11	
~~U-value~~U-factor										
Mass/7.0≤HC	0.097		0.158		0.158		0.097		0.158	
Other	0.050		0.076		0.076		0.076		0.076	
Windows										
~~U-value~~U-factor	~~0.72~~0.49		~~1.23~~0.81		~~1.23~~0.81		~~0.72~~0.49		~~0.72~~0.49	
Relative solar heat gain										
~~North~~	~~0.77~~		~~0.82~~		~~0.82~~		~~0.77~~		~~0.77~~	
~~Nonnorth~~	~~0.50~~		~~0.62~~		~~0.62~~		~~0.50~~		~~0.50~~	
	Non-North	North	Non-North	North	Non-North	North	Non-North	North	Non-North	North
0-10% WWR	0.49	0.72	0.61	0.61	0.61	0.61	0.47	0.61	0.46	0.61
11-20% WWR	0.43	0.49	0.55	0.61	0.61	0.61	0.36	0.51	0.36	0.51
21-30% WWR	0.43	0.47	0.41	0.61	0.39	0.61	0.36	0.47	0.36	0.47
31-40% WWR	0.43	0.47	0.41	0.61	0.34	0.61	0.31	0.47	0.31	0.40
Skylights										
~~U-value~~	~~0.85~~		~~1.31~~		~~1.31~~		~~0.85~~		~~0.85~~	
~~Solar heat gain coefficient~~										
~~Transparent~~	~~0.44~~		~~0.61~~		~~0.61~~		~~0.44~~		~~0.44~~	
~~Translucent~~	~~0.70~~		~~0.75~~		~~0.75~~		~~0.70~~		~~0.70~~	
U-factor Glass w/Curb	0.99		1.18		1.18		0.99		0.99	
Glass wo/Curb	0.57		0.68		0.68		0.57		0.57	
Plastic w/Curb	0.87		1.30		1.30		1.10		1.10	
SHGC Glass 0-2%	0.68		0.79		0.79		0.46		0.46	
2.1-5%	0.46		0.40		0.40		0.36		0.36	
SHGC Plastic 0-2%	0.77		0.79		0.77		0.77		0.71	
2.1-5%	0.58		0.65		0.62		0.62		0.58	

NURS39-Envelope (H25:X25)

The 1998-2007 vintage uses Title 24-2001, which does not include VT (visible transmittance) values, so these should not be shown. The cell formulas show them as based on SHGC x 1, but usually 1.05 is the most common conversion.

NURS40-Envelope (H13:H14, H18:H19)

These cells say "ASHRAE Handbook of Fundamentals for single pane & T24 2013 for double pane 33%-67%)" but no justification for using a 33%/67% weighting is provided. Other scorecards cite CBECS.

NURS41-Envelope (H13:X14, H18:X19)

Averaging the U-Factor and SHGC between single and double-pane options leads to non-physically possible windows. By weighting the resulting combination of U-Factor and SHGC, the resulting fenestration may not be physically possible and may cause problems with simulation engines such as EnergyPlus that expect a window to be built up from a specific pane of glass with specific properties. It might be better to have a mixture of single-pane and double-pane windows in a model (although that is a different compromise with reality). This is almost an exact repeat of the comment for the large hotel LHSC29-Envelope, small hotel SHSC39-Envelope, and office OFFM33-Envelope.

NURS42-Envelope (D13:D14, D18:D19, D23:D24)

These cells say "Single Pane," but the corresponding note in column H says "ASHRAE Handbook of Fundamentals for single pane & T24 2013 for double pane 33%-67%)" which means that it is not really just single pane.

NURS43-Envelope (n/a)

Several envelope table issues were reported for the small hotel that also apply to the hospital prototype table:

- SHSC32 – roof replacement should be weighted
- SHSC37 – use non-north for all windows
- SHSC37 – consider weighting all standards for historic values

For brevity, they will not be repeated.

NURS44-Envelope (I32:X36, E40:G43, Z30:AA32)

To verify the infiltration values, the four and six side areas were based on the values from the Prototype tab:

- Area 4 sides = 13 x 2 [height] x (430 + 140 + 430 + 140) [perimeter] = 29,640 ft2

- Area 6 sides = 29,640 + 2 x 430 x 140 = 150,040 ft2
- Ratio of 6 sides to 4 sides = 150040 / 56720 = 5.06

These values do match the values for 6-sided or 4-sided area in E40:G43 based on the values shown for area, height, or perimeter Z30:AA32. The area, height, and perimeter shown are inconsistent with the values shown on the prototype tab. There is uncertainty about those values as well, see NURS16-Prototype.

So, according to the referenced report

$$I_{design} \text{ at the building envelope } = 0.112 \times I_{75} \text{ at building envelope} \qquad 2$$

$$I_{design\ (4-side)} = I_{design\ (5-or\ 6-side)} \times \left(\frac{\text{Total leakage area } (5 - \text{or } 6 - \text{side})}{\text{Exterior Wall area}}\right) \qquad \textit{Equation 3}$$

Using these equations, the values should be:

Pre 1978	0.623 (not 0.485 as shown in scorecard)
1978-1997	0.623 (not 0.485 as shown in scorecard)
1998-2007	0.623 (not 0.485 as shown in scorecard)
2008-2018	0.567 (not 0.441 as shown in scorecard)
2019+	0.340 (not 0.264 as shown in scorecard)

These are quite far off. The error is due to the areas and size of the building.

NURS45-HVAC System (G4:G5)

For new construction, the source for the system type is "T24 2025 ACM 2025, Table 2 and 3," but in that document, Table 1 indicates that for "Healthcare Facilities," the standard design is "Same as the Proposed Design." The chosen systems, HP/DX with SZHP+DOAS for patient rooms and gas furnace/DX VAV for other zones, make sense, but the source should be something else.

NURS46-HVAC System (C3)

The heating system for existing vintages is shown as "Gas Furnace" in cell E3, but in the table below that shows for the 2008-2018 vintage of "COP=3 / Efficiency 80%", which describes a heat pump and a furnace.

NURS47-HVAC System (C8)

The heading of the column is "Gas Furnace," but two of the five rows show a heat pump. Perhaps the heading should simply be "heating system" instead, and the rows should indicate if a gas furnace or heat pump is used. Cell E8 is currently labeled "HP" and has a dash in row 12 where there is a heat pump. Overall, this table is confusing.

NURS48-HVAC System (D12)

The EER shown for the 2008-2018 vintage is 9.7, but the source Title 24-2013 shows EERs of 11.2, 11.0, 10.0, and 9.7 depending on the size, and this doesn't include a deduction of 0.2 for EERs with heating sections other than resistance heat.

NURS49-HVAC System (C12:G13)

The values and sources cited are not well enough explained to be verified. This covers 2008-2018 and new construction vintages for the efficiency of HVAC equipment. Since these are critical values, they need to be explained.

NURS50-HVAC System (C19:D20, C23:D24)

The total static pressure of 4.46 in w.c. for CAV does appear in PNNL 20405, 2010, Table 5.11, as shown below, but it is not clear that the range of 7,437 to 20,000 CFM is appropriate. Please also document which range is used. The source of the VAV 6.3.2 can also be found in the table, but it is not documented and corresponds to 4,648 to 20,000 CFM. The lack of documentation and assumptions also applies to the existing fan power assumption of 2.50 in w.c. for DOAS and 5.58 in w.c. for VAV.

Table 5.11. Rules for Establishing Fan Power Inputs for PI Analysis

System	CFM Range	TSP in. w.c.	Fan Efficiency
Fan Power Limitation Rule for 90.1-2004			
CAV	< 7,437	2.50	0.65
VAV	< 4,648	4.00	0.65
CAV	≥ 7,437 and < 20,000	4.46	0.65
CAV	≥ 20,000	4.09	0.65
VAV	≥ 4,648 and < 20,000	6.32	0.65
VAV	≥ 20,000	5.58	0.65
Unit Heater	All	0.2	0.65
PTAC, PTHP, or FCUs	All	1.33	0.65
Fan Power Limitation Rule for Addendum 90.1-04ac			
CAV	< 7,437	2.50	0.65
VAV	< 4,648	4.00	0.65
CAV	≥ 7,437	4.09	0.65
VAV	≥ 4,648	5.58	0.65
Unit Heater	All	0.2	0.65
PTAC, PTHP, or FCUs	All	1.33	0.65

NURS51-HVAC System (F23:24)

For new construction, a fixed value of 0.9 for the fan motor and drive efficiency is not correct, and instead, the size of the motor and ACM 2025 Table 13 should be used. This is described as the process for existing vintages but not for new construction. This is the same comment as LHSC52-HVAC System.

NURS52-HVAC System (G23:G24)

The new construction power per airflow of 0.209 W/cfm for DOAS and 1.013 W/cfm for VAV cites "ACM 2025, Table 14 and 15," but no additional explanation is provided. The 0.209 W/cfm does correspond to a value in Table 14 for "all other systems" for less than 5000 cfm, but it is not clear where these assumptions came from. Special filtering may be justified for a skilled nursing facility.

Table 14: Additional System Fan Power Allowance, in W/cfm by System Type

System No.	≤ 5,000 cfm	> 5,000 cfm; ≤ 10,000 cfm	> 10,000 cfm
MZ-VAV (Systems 5 and 6)	0.205	0.174	0.159

System No.	≤ 5,000 cfm	> 5,000 cfm; ≤ 10,000 cfm	> 10,000 cfm
All other (Systems 1, 3, 7, and 9)	0.209	0.182	0.162

Table 15: Adjustment Credits (Multizone VAV) (W/cfm)

Device	≤ 5,000 cfm	> 5,000 cfm; ≤ 10,000 cfm	> 10,000 cfm
Return of exhaust systems required by code to be fully ducted	0.089	0.100	0.116
Exhaust filters, scrubbers, or other exhaust treatment (calculation required, see note)	0.177	0.198	0.231
Particulate filtration credit: MERV 16 or greater and	0.265	0.280	0.333

Device	≤ 5,000 cfm	> 5,000 cfm; ≤ 10,000 cfm	> 10,000 cfm
electronically enhanced filters			
Carbon and other gas-phase air cleaners (calculation required, see note)	0.176	0.188	0.224
Biosafety cabinet (calculation required, see note)	0.177	0.198	0.231
Energy Recovery (included only if standard design requires heat recovery)	0.374	0.318	0.289

NURS53-Ventilation (E2:E65)

This column is described as "cfm/person [T24, 2025, Equation 120.1-F and Table 120.1-A, and ASHRAE 170 Table 9.1, ASHRAE 62.1 User's Manual Equation 6-1]" but the values are calculated based on the column "Total cfm Driven by Area" divided by the number of people. It does not seem to be based on Title 24 or ASHRAE 170 at all. I would recommend that this column only include values that are from a source that uses cfm/person. If a calculated cfm/person value is desired, a different column should include that, perhaps after the Total cfm column.

NURS54-Ventilation (A2:A65)

The list of "Space/Thermal Zone" on the Ventilation tab is not the same as the list on the Zones tab. The version on the Ventilation tab has 63 items, and on the Zones tab, it has 113. It may be that some zones that are very similar (such as patient rooms) are not replicated, but it is confusing what exactly is going on, and if other spaces are missing.

NURS55-Ventilation (F2:F65)

Column F is "cfm/ft2" and described as:

“[[T24, 2025, Equation 120.1-F and Table 120.1-A, and ASHRAE 170 Table 9.1, ASHRAE 62.1 User’s Manual Equation 6-1]”

For requirements that are in cfm/ft2, it makes sense to include them in the column, and these include the Title 24-2025 values from Table 120.1-A, but it is confusing that a requirement that is expressed in ACH (air changes per hour) is recalculated here in a formula. Instead, I would recommend adding a new column for requirements that are expressed as ACH, such as ASHRAE 170 Table 9.1.

NURS56-Ventilation (F21:F22)

Why should "Special_purpose_N_1" room and "Special_purpose_S_1" have a 2 ACH requirement? It is not clear what space type it is being mapped to.

NURS57-Ventilation (L2:L65)

Please include in the notes for each row the following:

- the source of the required value, including the name of the code or standard
- the table number being referenced for that specific value
- the space category from the referenced table

It is difficult to track the source of each value without specific information, and we are trying to document each assumption. If numbers for a row come from multiple sources, then the note should specify which value came from which source, and each source should be separately referenced. Simply saying that all the numbers come from all the sources makes it difficult to confidently check any particular number.

NURS58-Ventilation (F26, F20, F15)

Why does Bathroom_Patient_1 have a 0.3 cfm/ft2 ventilation rate (based on 2 ACH) while Bathroom/Restroom_1 and Restroom_Dietic_Services_1 have 0.15 cfm/ft2?

NURS59-Ventilation (F28)

Why is Pharmacy_1 using 0.15 cfm/ft2, which is based on Title 24-2025 Table 120.1A, when Standard 170 has pharmacies based on 2 ACH, which would be 0.3 cfm/ft. How is this decision being made?

NURS60-Ventilation (multiple in column F)

Why do some storage rooms have 0.15 cfm/ft2

- Storage_Admin_1
- Storage_Large_1
- Storage_Service_CW_2
- Storage_Service_CE_2

And others have 0 cfm/ft2

- Storage_Dietic_Services_1
- Storage_Office_Dietic_Services_1
- Specialized_Storage_1
- Wheelchair_Storage_1
- Specialized_Storage_2
- Linen_Storage_2
- Supply_Storage_2

NURS61-Ventilation (F38, F58)

Why does Corridor_1 have 0.15 cfm/ft2 while Corridor_2 has 0.3 cfm/fts (based on 2 ACH)?

NURS62-Ventilation (multiple in column F)

Patient rooms seem to have different ventilation rates. These have 0.3 cfm/ft2 (based on 2 ACH)

- Patient_Room_N_1
- Patient_Room_W_1
- Patient_Room_S_1
- Patient_Room_E_2

These have 0.15 cfm/ft2

- Patient_Room_S_2

These have 0.6 cfm/ft2 (based on 4 ACH)

- Patient_Room_N_2
- Patient_Room_W_2

NURS63-Ventilation (F36, F53)

Nursing_Station_1 has zero ventilation, and Nursing_Station_2 has 0.15 cfm/ft2. It may be more appropriate to use 0.3 cfm/ft2 (based on 2 ACH from ASHRAE Standard 170)

NURS64-Ventilation (Column G)

For the exhaust rates, it is not clear where the 0.016, 0.7, 0.5, 3, and 1 values are from, and if they are correct. The 0.016 values correspond to restrooms and have a formula of 20/1250, but that does not clarify the origin of the value and how it was computed.

NURS65-Ventilation (n/a)

The remaining portions of the ventilation rate table should be examined closely due to the number of errors found.

NURS66-Ventilation (E70, B81:C82)

The hood length calculation of 16 hoods, 2.5 ft in length each, should result in 40 ft, not 41 ft as shown in the table. Adding 1 foot more for a buffer would be 41 ft, not 42 ft.

NURS67-Ventilation (B76)

What is the basis for the count of 16 hoods? Each one is assumed to be 2.5 ft by 1.75 ft.

NURS68-Ventilation (E69)

The hood length has a reference of "CKV, 2004," but no explanation is provided of what that source is.

NURS69-Thermostat (all)

The sources of the thermostat setpoints shown are not clear, especially the reference to OSHPD 2, which is not defined. The use of a cooling setpoint that is below a heating setpoint is quite unusual. For patient areas, the heating setback and cooling setback that are equivalent to the cooling setpoint and heating setpoint, respectively, imply that they are simply defining the deadband.

Thermostat Setpoints - Skilled Nursing Facility																		
Space/Thermal Zone	Setpoint	CZ1	CZ2	CZ3	CZ4	CZ5	CZ6	CZ7	CZ8	CZ9	CZ10	CZ11	CZ12	CZ13	CZ14	CZ15	CZ16	Source
Patient Areas	Heating Setpoint (F)	75	75	75	75	75	75	75	75	75	75	75	75	75	75	75	75	T24 2022 - Appendix 5.4B , OSHPD 2, ASHRAE 170, AHRAE 55
	Heating Setback (F)	71	71	71	71	71	71	71	71	71	71	71	71	71	71	71	71	
	Cooling Setpoint (F)	71	71	71	71	71	71	71	71	71	71	71	71	71	71	71	71	
	Cooling Setback (F)	75	75	75	75	75	75	75	75	75	75	75	75	75	75	75	75	
Non-Patient Areas	Heating Setpoint (F)	75	75	75	75	75	75	75	75	75	75	75	75	75	75	75	75	T24 2022 - Appendix 5.4B , OSHPD 2, ASHRAE 170, AHRAE 55
	Heating Setback (F)	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70	70	
	Cooling Setpoint (F)	75	75	75	75	75	75	75	75	75	75	75	75	75	75	75	75	
	Cooling Setback (F)	85	85	85	85	85	85	85	85	85	85	85	85	85	85	85	85	

ASHRAE Standard 170 lists recommended filter efficiencies for air systems serving specific nursing home areas Table 9.1.
2019 ASHRAE Handbook—HVAC Applications Chapter 9 Winter DD for patient areas 75 and 70 for non-patient area. Summer DD 75 for all and 60% rh
OSHPD 2: 71-81
Title 24 Part 4 Does not allow less than 60F setback for SNF
Patient Rooms. Each patient room should have individual temperature control. Air pressure in general patient suites can be neutral in relation to other areas. Most governmental design criteria and codes require that all air from toilet rooms be exhausted directly outdoors.

It appears that the ASHRAE Handbook Applications Chapter 9 is the intended source of values and says:

ASHRAE *Standard* 170 lists recommended filter efficiencies for air systems serving specific nursing home areas and recommended minimum ventilation rates and desired pressure relationships. Recommended interior winter design temperature is 75°F for areas occupied by patients and 70°F for nonpatient areas. Provisions to maintain minimum humidity levels in winter depend on the severity of the climate and are best left to the designer's judgment. Where air conditioning is provided, the recommended interior summer design temperature and humidity is 75°F and a maximum of 60% relative humidity. Temperature should be controlled in individual rooms. In areas with severe climates, patient rooms may have supplementary heat along exposed walls. In moderate climates (i.e., where outdoor winter design conditions are 30°F or above), overhead heating may be used.

But then the patient area cooling setpoint should be 75F and the non-patient heating setpoint should be 70F.

NURS70-Thermostat (n/a)

Since individual control is expected for patient rooms, the thermostat value assumptions are a complicated topic. Using a uniform setpoint for all patient rooms is a simple approach, but I would recommend a more comprehensive approach for the prototype. Some considerations to include would be:

- Diversity of temperatures that patients find comfortable
- Comfortable temperatures are likely different during the day and when sleeping
- Patients do not pay extra for energy use to obtain greater comfort
- Patients who are older are more likely to desire warmer temperatures

To try to capture these effects, use several different schedules for patient rooms that have a wide diversity of values, including deep setbacks for unoccupied rooms. This is a similar comment to LHSC56-Thermostat and SHSC53-Thermostat.

NURS71-Water Heater (various)

It is not clear what space type from the source "24, 2022, Appendix 5.4A, sheet: 2022 SpaceFuncData-Input" was used for the following spaces, the water heating type and gal/hr-person is shown in parentheses:

- Special_purpose_N_1 (electric 0.240 gal/hr-person)
- Special_purpose_S_1 (electric 0.240 gal/hr-person)
- Clean_Utility_1 (0.180 gal/hr-person)
- Soiled_Utility_1 (0.180 gal/hr-person)
- Recreation_1 (0.180 gal/hr-person)
- Clean_Utility_2 (0.180 gal/hr-person)
- Soiled_Utility_2 (0.180 gal/hr-person)
- Recreation_CW_2 (gas 0.180 gal/hr-person)
- Recreation_CE_2 (gas 0.180 gal/hr-person)

NURS72-Water Heater (various)

For the following spaces, which have positive gal/hr-person, do not specify the type of water heater:

- Clean_Utility_1 (0.180 gal/hr-person)
- Soiled_Utility_1 (0.180 gal/hr-person)
- Clean_Utility_2 (0.180 gal/hr-person)

- Soiled_Utility_2 (0.180 gal/hr-person)

NURS73-Water Heater (A65:B65, E65)

Corridor_2 is shown with 0.578 gal/hr-person and no specified water heater type. This looks like a copy-paste error since other corridors show the correct assumption of 0.0 gal/hr-person.

NURS74-Water Heater (column F)

Most of column F, "Peak Hot Water Usage," is computed based on the number of people times the gal/hr-person, but in a few rows related to bathrooms, a value has been added manually. The value 64.8 gal/hr is based on the note "CalGreen Section 5.303.3.2 =1.8 gpm per shower with 60% hot water."

5.303.3.3.1 Single showerhead.

Showerheads shall have a maximum flow rate of not more than 1.8 gallons per minute at 80 psi. Showerheads shall be certified to the performance criteria of the US EPA WaterSense Specification for Showerheads.

This computation should probably be in a different column of the spreadsheet to reduce confusion.

NURS75-Water Heater (B79:B82)

The peak hot water usage total by water heater fuel type uses the maximum of a column instead of summing it. I believe this formula is wrong since it essentially reflects the use of the Bathroom_Patient_2 only. Why is "Gas-existing" simply pointing to Total, which is the sum of "Gas storage WH" and "Electric resistance WH"? It doesn't seem to be logical. The total line, not reflecting the three items above, is also very misleading.

NURS76-Water Heater (B86)

The cold-water supply temperature of 55°F shown in the scorecard is consistent with the citation of the ACM 5.9.2 Rated Capacity, but better data is available that may be closer to how water heaters are actually sized using the local water mains temperature. The Appendix 5.4B (called CEC-400-2022-009-CMF-REV2-AP5.4B.xlsx) includes a WaterMain tab and WaterMainMonth tab that show climate-specific values for the entering water

temperature. The lowest of these values for each climate zone is more likely to be used for sizing water heaters; the monthly values vary from 41.4 in CZ16 to 63.2 in CZ15. Using these values would have a large impact on the sizing of the water heater. This is the same comment as OFFM52-Water Heater.

NURS77-Water Heater (F79:F81)

The storage volume calculation is the formula "=ROUNDUP(24*(B79/B82),0)" where B79 and B82 are the Gas storage WH peak hot water use and the Total peak hot water use. This formula essentially says that the storage volume is some fraction of 24 based on how much is gas versus total. This means that the value cannot be larger than 24 gallons, which does not make sense for buildings with larger hot water needs, such as this one. This is smaller than a single 40-gallon water heater that a single-family home usually has.

NURS78-Water Heater (Below Line 113)

Good information is provided about the use of water based on a table from the California Plumbing Code for healthcare facilities, but it doesn't seem to be getting used in the calculations.

NURS79-Water Heater (n/a)

Laundry and special sanitation typically would require additional water heaters to provide higher temperature water or perhaps boilers for providing steam. Similar to comment HOSP36.

NURS80-Interior Lights (column G)

The use of Title 24-2016 for all existing vintages is not consistent with assumptions used for other buildings, and no specific justification for this has been cited. Other buildings used:

- Hospital – Title 24-2019
- Medium office - Title 24-2013
- Warehouse - Title-24 2016
- HotelLarge – Title 24-2019
- HotelSmall – Title 24-2019

No justification is used for using Title 24-2016 for existing lighting power densities, and the selection used for different prototype scorecards seems arbitrary. In general, assuming that all lighting systems would be replaced in all existing vintages of nursing homes by 2016 seems optimistic. Instead, perhaps a weighted value based on multiple versions of Title 24 would make sense.

NURS81-Interior Lights (column B)

For new construction in a skilled nursing facility, the lighting power for "Aging Eye/Low-vision," from T24, 2025, Table 140.6-C should be used, even if not exclusively serving aging populations.

TABLE 140.6-C AREA CATEGORY METHOD - LIGHTING POWER DENSITY VALUES (WATTS/FT²)

Building Type/Use	Primary Function Area	Allowed Lighting Power Density for General Lighting (W/ft²)	Additional Lighting Power Qualified Lighting Systems	Additional Lighting Power Additional Allowance [3] (W/ft², unless noted otherwise)
Aging Eye/Low-vision[11]	Corridor Area	0.70	Decorative/Display	0.30
Aging Eye/Low-vision[11]	Dining	0.80	Decorative/Display	0.30
Aging Eye/Low-vision[11]	NA	0.80	Tunable white or dim-to-warm[10]	0.10
Aging Eye/Low-vision[11]	Lobby, Main Entry	0.85	Decorative/Display	0.30
Aging Eye/Low-vision[11]	Lobby, Main Entry	0.85	Transition Lighting OFF at night[12]	0.95
Aging Eye/Low-vision[11]	Lobby, Main Entry	0.85	Tunable white or dim-to-warm[10]	0.10
Aging Eye/Low-vision[11]	Lounge/Waiting Area	0.80	Decorative/Display	0.30
Aging Eye/Low-vision[11]	Lounge/Waiting Area	0.80	Tunable white or dim-to-warm[10]	0.10
Aging Eye/Low-vision[11]	Multipurpose Room	0.85	Decorative/Display	0.30
Aging Eye/Low-vision[11]	Multipurpose Room	0.85	Tunable white or dim-to-warm[10]	0.10
Aging Eye/Low-vision[11]	Religious Worship Area	1.00	Decorative/Display	0.30
Aging Eye/Low-vision[11]	Religious Worship Area	1.00	Tunable white or dim-to-warm[10]	0.10
Aging Eye/Low-vision[11]	Restroom	1.00	Decorative/Display	0.20
Aging Eye/Low-vision[11]	Stairwell	0.80	Decorative/Display	0.30

These are often 0.2 W/sqft higher than the non-aging eye/low-vision values. The footnote related to this is shown below:

11. Aging Eye/Low-vision areas can be documented as being designed to comply with the light levels in ANSI/IES RP-28 and are or will be licensed by local or state authorities for either senior long-term care, adult day care, senior support, and/or people with special visual needs.

And RP-28 and RP-29 are shown as footnotes on the Interior Lighting tab of the scorecard under the table. But the values shown in the table do not reflect this, including:

- Lobby_1
- Stairway_Elevator_1
- Bathroom/Restroom_1
- Bathroom_Patient_1

- Recreation_1
- Resident_Dining_1
- Corridor_1
- Vestibule_1
- Bathroom/Restroom_2
- Bathroom_Patient_2
- Recreation_CW_2 (presuming this is a multipurpose room)
- Recreation_CE_2 (presuming this is a multipurpose room)
- Resident_Dining_CW_2
- Resident_Dining_CE_2
- Corridor_2
- Stairway_Elevator_2

Stairwells and dining, the additional lighting power is 0.30 W/ft2 (not 0.35 W/ft2 for non-aging eye). For corridors, the additional lighting power is 0.30 W/ft2 (not 0.25 W/ft2 for non-aging eye)

NURS82-Interior Lights (B11, B15, B28)

The following spaces show a new construction LPD of 0.455 W/ft2, but it should be 0.45.

- Staff_Dining_1
- Lockers_Staff_1
- Laundary_1

NURS83-Interior Lights (column B)

Storage spaces in new construction should have an LPD of 0.40 W/ft2, but the following spaces have either 0.455 or 0.45:

- Storage_Admin_1
- Staff_Dining_1
- Storage_Dietic_Services_1
- Storage_Office_Dietic_Services_1
- Storage_Large_1
- Specialized_Storage_1
- Housekeeping_Room_1
- Wheelchair_Storage_1

- Recreation_Equipment_1
- Storage_Service_CW_2
- Storage_Service_CE_2
- Recreation_Equipment_2
- Specialized_Storage_2
- Linen_Storage_2
- Supply_Storage_2

NURS84-Interior Lights (A18:A19)

The space: Stairway_Elevator_1 and Stairway_Elevetor_1 is shown twice, but I think the second is simply a misspelling.

NURS85-Interior Lights (B36, B54)

Nurses' stations for new construction are shown with an LPD of 0.75 W/ft2, but should be 0.85 W/ft2.

NURS86-Interior Lights (B33, B35, B48, B49)

The source of the LPD for new construction for the following spaces is not clear:

- Clean_Utility_1
- Soiled_Utility_1
- Clean_Utility_2
- Soiled_Utility_2

The LPD value shown is 0.55 W/ft2, which may correspond to a Healthcare Facility and Hospital Medical Supply Room, but it might be more appropriate to just consider them as laundry areas, which have an LPD of 0.45 W/ft2

NURS87-Interior Lights (B33, B35, B48, B49)

For the existing vintage, the LPD for the following laundry-related spaces should be 0.7 W/ft2, not 0.6 W/ft2

- Clean_Utility_1

- Soiled_Utility_1
- Clean_Utility_2
- Soiled_Utility_2

NURS88-Interior Lights (B22:C23, E22:23)

The source of the LPD of 0.85 W/ft2 for new construction for Special_purpose_N_1 and Special_purpose_S_1 is not clear. Neither is the assumption used for existing construction using 1.2 W/ft2. Please add notes to explain this.

NURS89-Interior Lights (C3, F3)

The heading of the column for new construction and existing vintages of "Additional LPD for Portable Lighting (W/ft^2)" should probably say "Additional Lighting Power Additional Allowance" since the additional allowances are not related to them being portable.

NURS90-Interior Lights (n/a)

It would be easier to understand and review if another column were added for both new construction and existing that showed the "Building Type/Use" and "Primary Function Area" from the referenced codes that were assumed.

NURS91-Interior Lights (E39)

The Corridor_1 LPD of 0.0 W/ft2 for existing should be changed to 0.6 W/ft2 to match the code.

NURS92-Interior Lights (E37, E55, E56)

The recreation spaces shown below for existing have an LPD of 1.3 W/ft2, but the only primary function area with that LPD is "Civic Meeting Place Area," which seems like an odd choice for the following rooms.

- Recreation_1
- Recreation_CW_2
- Recreation_CE_2

NURS93-Interior Lights (F4, F11, F38, F57, F58)

Based on Title 24-2016 Table 140.6C footnote 3, additional lighting power is allowed for ornamental lighting of 0.5 W/ft, which covers:

- Lobby_1
- Staff_Dining_1
- Resident_Dining_1
- Resident_Dining_CW_2
- Resident_Dining_CE_2

These spaces are likely to take advantage of this credit.

NURS94-Ext Lights (C11)

The parking area of 10,000 ft2 has no justification based on the number of workers, visitors, or patients.

NURS95-Ext Lights (C13)

Assuming zero hardscape ornamental lighting (and no outdoor path lighting) presumes no outdoor spaces for patients that are available after dark and the facility is not intended to look good at night. This seems unlikely.

NURS96-Ext Lights (C14)

The 570 ft perimeter used by the formula to determine the building façade lighting is inconsistent with the description of the building being 430 ft x 140 ft shown on the prototype tab in cell C8.

NURS97-Ext Lights (A15:A20)

The exterior lighting table should not include these items that are not appropriate for a nursing home, even using zero values.

- Drive-up windows

- Sales canopy
- Non-sales canopy
- Student Pick up/drop off zone
- Security lighting for retail parking

NURS98-Ext Lights (n/a)

Where are the exterior lighting assumptions for the vintage models? One assumes that those should be higher. This is the same comment as LHSC72-Ext Lights, SHSC67-Ext Lights, and HOSP44-Ext Lights.

NURS99-Ext Lights (B14)

The building façade area assumes a factor of 0.5 of the perimeter walls; less than two stories are lit, but Table 36 from the CASE report suggests using 5% for "other" buildings that are not on the list. This is the same comment as HOSP46-Ext Lights.

NURS100-Equipment (B29, B30, B31, B45, B46, B49, B50, B51, B52)

The plug load power density for storage should be 0.2 W/ft2 and appears as that for some spaces but not for the following, which show 0.0 W/ft2:

- Specialized_Storage_1
- Housekeeping_Room_1
- Wheelchair_Storage_1
- Recreation_Equipment_1
- Storage_Service_CW_2
- Storage_Service_CE_2
- Recreation_Equipment_2
- Specialized_Storage_2
- Linen_Storage_2
- Supply_Storage_2

NURS101-Equipment (E3:E65)

The source is shown as “T24 2025 - Appendix 5.4A: 2022 SpaceFuncData-Input," but this is actually from Title 24-2022 since this appendix is not yet available for Title 24-2025.

NURS102-Equipment (G1:AE6)

The refrigeration schedule for all zones should be located and described on the Schedules tab, and a source, even if professional judgment, should be cited.

NURS103-Equipment (B21:D22, B32:D32, B34:D34, B36:D36, B45:D48, B54:D55)

Please describe the assumed "2022 NACM Space Type" for the spaces:

- Special_purpose_N_1
- Special_purpose_S_1
- Clean_Utility_1
- Soiled_Utility_1
- Recreation_1
- Clean_Utility_2
- Soiled_Utility_2
- Recreation_CW_2
- Recreation_CE_2

And describe the reason for those choices.

NURS104-Equipment (n/a)

No electric load for the elevators is described on the equipment tab yet; elevators are crucial in a skilled nursing facility with more than one floor, like this prototype. In addition, they are indicated as present in the list of zones and on the layouts on the Zones tab. These will be larger elevators since serving a facility where gurneys will be transported which imply higher power motors and more elevator lighting and ventilation.

NURS105-Schedules (X16:Z18, X46:Z48)

Corridor occupancy averages about 30% for most hours, but from 8 pm to 10 pm every day goes to 80% to 90%. This is repeated for corridor plug loads during these hours. No explanation for this is provided.

NURS106-Schedules (Column B)

Column B describes the source for each of the 40 schedules shown on the tab, but unfortunately, every description is the same for these 40 schedules as shown below:

> T24 2022 - Appendix 5.4B
> OSHPD 2
> FGI 2022
> ASHRAE 170
> ASHRAE 90.1 (Appendix G)
> 2019 ASHRAE Handbook, HVAC Applications CHAPTER 9, HEALTH CARE FACILITIES

This means that either every schedule is from all of these sources, which provide them consistently (which seems extremely unlikely), or that the source of an individual schedule is one of these. In this second case, as a reviewer, I would need to look at all of these sources and at any scheduled referenced and see if they match in order to confirm them.

Please replace this generic, unspecific list of sources with a specific source for each schedule, and please also be specific in exactly which schedule is used, since several of these sources have multiple possible schedules in them. For sources without schedules but that have text that implies a certain type of schedule, please list the section and paragraph that describes the reason for the assumption. Most of these sources are dozens to hundreds of pages.

The T24 2022 - Appendix 5.4B spreadsheet should be used as the primary source, and other sources should be used only with explanation. If the Appendix 5.4B spreadsheet is used, the specific tab of the 14 tabs should be disclosed. I tried using this as a source and fitting together the schedules that seemed to be closest, but it is not worth guessing and using trial and error to figure out what the assumptions were, since the author of the spreadsheet already knows the assumptions used and simply did not disclose them.

It is nearly impossible to review this tab without a reasonable citation of the sources of information. This tab was not reviewed further based on this omission.

NURS107-Schedules (C4:AB6)

In long-term or short-term care facilities, one would expect that the patient rooms are only fully occupied at night and are less occupied during the day. And common areas should be more occupied during the day. The schedules shown for “Occupant Zone” show 100% for

all hours. It would make more sense to have finer grained schedules assigned to spaces to account for this with different schedules for patient rooms and spaces that patients occupy during the day.

NURS108-Weights (general)

Some of the information about the number of beds and area of the facility per bed that appears on the "Weights" tab may be more consistent with other scorecards and appear on the Notes or Prototype tab. In addition, the source for the data in the chart should be indicated, and for the slide and plan view of the facility.